

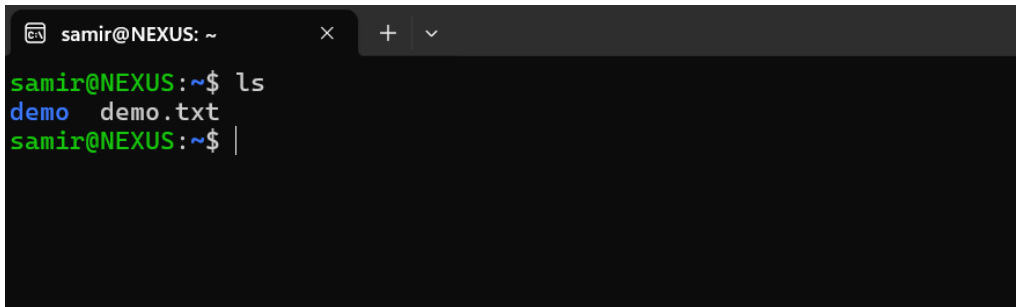
Lab 1: Basic Linux Commands

A. Basic Directory Navigation Commands

1. Command: `ls`

Interpretation: List information about files and directories in the current directory.

Output:

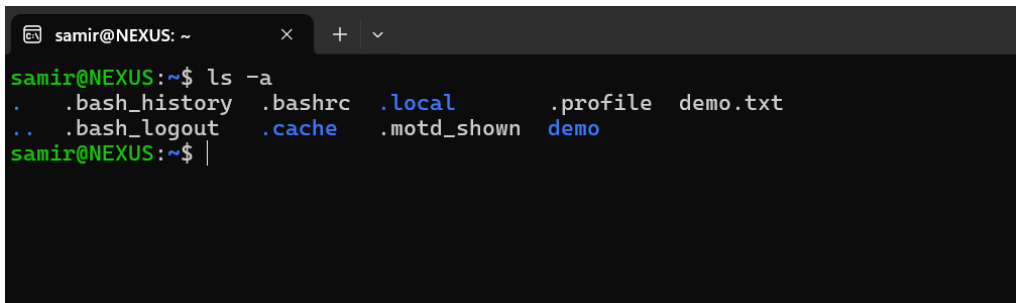
A terminal window titled 'samir@NEXUS: ~' with a dark background. The prompt is 'samir@NEXUS:~\$'. The command 'ls' has been entered, and the output is 'demo demo.txt'. The cursor is on a new line.

```
samir@NEXUS:~$ ls
demo demo.txt
samir@NEXUS:~$ |
```

2. Command: `ls -a`

Interpretation: List all contents of the current directory, including hidden files and directories (those starting with '.').

Output:

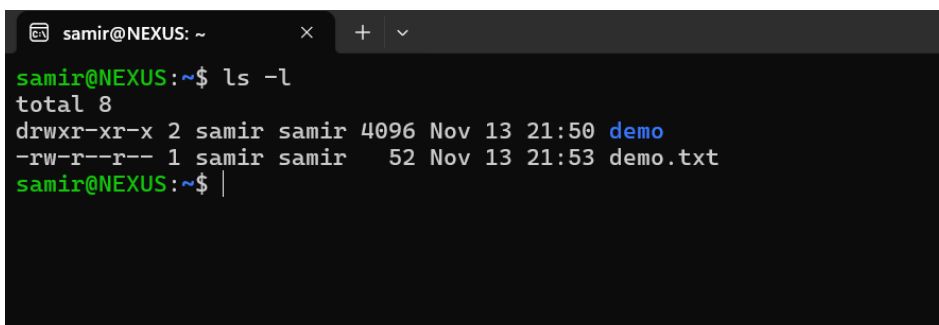
A terminal window titled 'samir@NEXUS: ~' with a dark background. The prompt is 'samir@NEXUS:~\$'. The command 'ls -a' has been entered, and the output lists hidden files and directories: '. .bash_history .bashrc .local .profile demo.txt', '.. .bash_logout .cache .motd_shown demo'. The cursor is on a new line.

```
samir@NEXUS:~$ ls -a
. .bash_history .bashrc .local .profile demo.txt
.. .bash_logout .cache .motd_shown demo
samir@NEXUS:~$ |
```

3. Command: `ls -l`

Interpretation: Display contents in long listing format, showing permissions, number of links, owner, group, size, and timestamp.

Output:

A terminal window titled 'samir@NEXUS: ~' with a dark background. The prompt is 'samir@NEXUS:~\$'. The command 'ls -l' has been entered, and the output shows detailed file information: 'total 8', 'drwxr-xr-x 2 samir samir 4096 Nov 13 21:50 demo', '-rw-r--r-- 1 samir samir 52 Nov 13 21:53 demo.txt'. The cursor is on a new line.

```
samir@NEXUS:~$ ls -l
total 8
drwxr-xr-x 2 samir samir 4096 Nov 13 21:50 demo
-rw-r--r-- 1 samir samir 52 Nov 13 21:53 demo.txt
samir@NEXUS:~$ |
```

4. Command: ls -lh

Interpretation: Display contents in long listing format with file sizes in human-readable format (KB, MB, GB).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ ls -lh  
total 8.0K  
drwxr-xr-x 2 samir samir 4.0K Nov 13 21:50 demo  
-rw-r--r-- 1 samir samir 52 Nov 13 21:53 demo.txt  
samir@NEXUS:~$ |
```

5.Command: ls -R

Interpretation: Recursively display contents of the current directory and all its subdirectories.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ ls -R  
.:  
demo demo.txt  
  
./demo:  
demo1.txt demo2.txt demo3.txt demo4.pdf demo5.html hey  
  
./demo/hey:  
hey.mp4 hey.png hey.py hey.xml  
samir@NEXUS:~$ |
```

6. Command: ls -ltr

Interpretation: List contents in long format (-l), sorted by time (-t) in reverse order (-r), showing oldest files first.

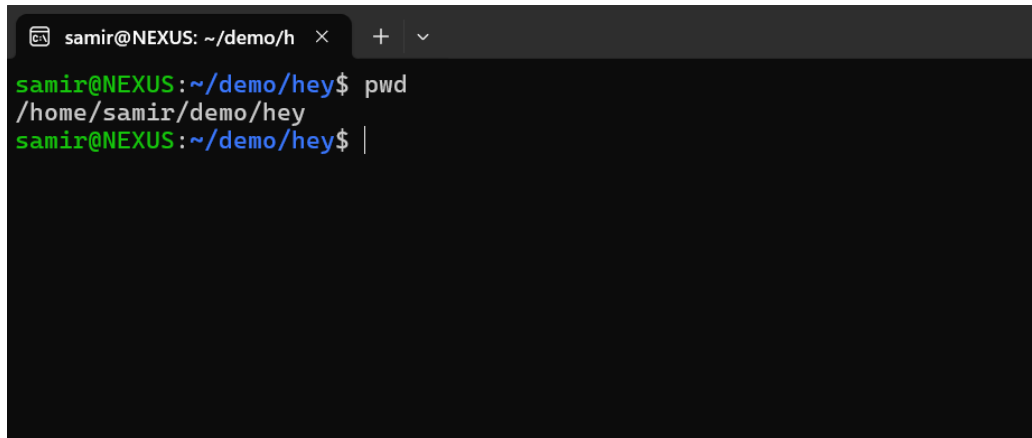
Output:

```
samir@NEXUS: ~/demo/h  
samir@NEXUS:~/demo/hey$ ls -ltr  
total 0  
-rw-r--r-- 1 samir samir 0 Nov 13 22:03 hey.mp4  
-rw-r--r-- 1 samir samir 0 Nov 13 22:03 hey.py  
-rw-r--r-- 1 samir samir 0 Nov 13 22:03 hey.png  
-rw-r--r-- 1 samir samir 0 Nov 13 22:03 hey.xml  
samir@NEXUS:~/demo/hey$ |
```

7.Command: pwd

Interpretation: Print the current working directory's full path.

Output:

A terminal window with a dark background. The title bar shows 'samir@NEXUS: ~/demo/h' and window controls. The prompt is 'samir@NEXUS: ~/demo/hey\$'. The command 'pwd' is entered. The output is '/home/samir/demo/hey'. The prompt is now 'samir@NEXUS: ~/demo/hey\$' with a cursor.

```
samir@NEXUS: ~/demo/h × + v
samir@NEXUS: ~/demo/hey$ pwd
/home/samir/demo/hey
samir@NEXUS: ~/demo/hey$ |
```

8.Command: clear

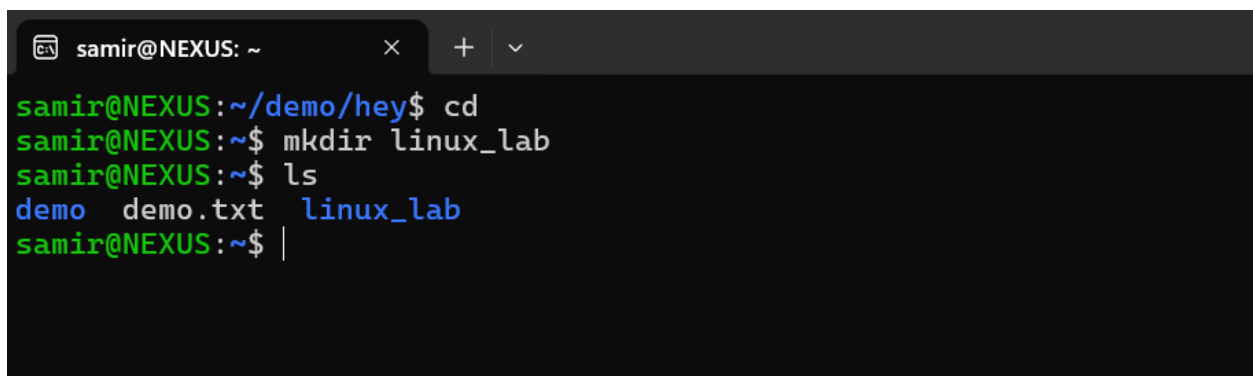
Interpretation: Clear the terminal screen.

B. Directory Creation and Navigation

9. Command: mkdir linux_lab

Interpretation: Create a new directory named 'linux_lab'.

Output:

A terminal window with a dark background. The title bar shows 'samir@NEXUS: ~' and window controls. The prompt is 'samir@NEXUS: ~/demo/hey\$'. The command 'cd' is entered. The prompt changes to 'samir@NEXUS: ~\$'. The command 'mkdir linux_lab' is entered. The prompt is 'samir@NEXUS: ~\$'. The command 'ls' is entered. The output is 'demo demo.txt linux_lab'. The prompt is 'samir@NEXUS: ~\$' with a cursor.

```
samir@NEXUS: ~ × + v
samir@NEXUS: ~/demo/hey$ cd
samir@NEXUS: ~$ mkdir linux_lab
samir@NEXUS: ~$ ls
demo demo.txt linux_lab
samir@NEXUS: ~$ |
```

10.Command: cd linux_lab

Interpretation: Change current directory to 'linux_lab'.

Output:

```
samir@NEXUS: ~/linux_lal x + v
samir@NEXUS:~$ cd linux_lab
samir@NEXUS:~/linux_lab$ |
```

11. Command: `cd ..`

Interpretation: Move up one directory level (to parent directory).

Output:

```
samir@NEXUS: ~ x + v
samir@NEXUS:~/linux_lab/lab1$ cd ..
samir@NEXUS:~/linux_lab$ cd lab1
samir@NEXUS:~/linux_lab/lab1$ cd
samir@NEXUS:~$ |
```

C. File Creation and Manipulation

12. Command: `touch file1.txt`

Interpretation: Create an empty file named 'file1.txt'.

Output:

```
samir@NEXUS: ~/linux_lal x + v
samir@NEXUS:~$ cd linux_lab
samir@NEXUS:~/linux_lab$ ls
lab1
samir@NEXUS:~/linux_lab$ cd lab1
samir@NEXUS:~/linux_lab/lab1$ touch file1.txt
samir@NEXUS:~/linux_lab/lab1$ ls
file1.txt
samir@NEXUS:~/linux_lab/lab1$ |
```

13. Command: `touch doc1.txt doc2.txt doc3.txt`

Interpretation: Create multiple empty files named 'doc1.txt', 'doc2.txt', and 'doc3.txt' simultaneously.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾
samir@NEXUS:~/linux_lab/lab1$ touch doc1.txt doc2.txt doc3.txt
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt file1.txt
samir@NEXUS:~/linux_lab/lab1$ |
```

14. Command: nano file1.txt

Interpretation: Open file1.txt in the nano text editor for editing.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾ - □ ×
GNU nano 6.2 file1.txt
hey folks, Are you learning linux commands??

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

15. Command: cat file1.txt

Interpretation: Display the contents of file1.txt in the terminal.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾
samir@NEXUS:~/linux_lab/lab1$ nano file1.txt
samir@NEXUS:~/linux_lab/lab1$ cat file1.txt
hey folks, Are you learning linux commands??
samir@NEXUS:~/linux_lab/lab1$ |
```

D. File and Directory Operations

16. Command: cp file1.txt file2.txt

Interpretation: Copy the contents of file1.txt to file2.txt. If file2.txt doesn't exist, it will be created.

Output:

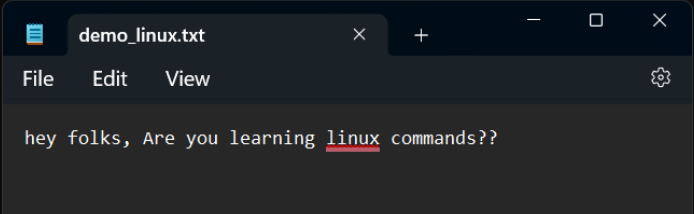
```
samir@NEXUS: ~/linux_lab x + v
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt file1.txt
samir@NEXUS:~/linux_lab/lab1$ cp file1.txt file2.txt
samir@NEXUS:~/linux_lab/lab1$ cat file2.txt
hey folks, Are you learning linux commands??
samir@NEXUS:~/linux_lab/lab1$ |
```

17. Command: `cp file1.txt /mnt/c/Users/samir/Desktop/demo_linux.txt`

Interpretation: Copy file1.txt content to the Desktop demo_linux.txt.

Output:

```
samir@NEXUS: ~/linux_lab x + v
samir@NEXUS:~/linux_lab/lab1$ cp file1.txt /mnt/c/Users/samir/Desktop/demo_linux.txt
samir@NEXUS:~/linux_lab/lab1$
```



18. Command: `mv file1.txt are_you_learning.txt`

Interpretation: Rename file1.txt to are_you_learning.txt.

Output:

```
samir@NEXUS: ~/linux_lab x + v
samir@NEXUS:~/linux_lab/lab1$ mv file1.txt are_you_learning.txt
samir@NEXUS:~/linux_lab/lab1$ ls
are_you_learning.txt doc1.txt doc2.txt doc3.txt file2.txt
samir@NEXUS:~/linux_lab/lab1$ |
```

19. Command: `mv file1.txt /mnt/c/Users/samir/Desktop/`

Interpretation: Move file1.txt to the Documents directory.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾
samir@NEXUS:~/linux_lab/lab1$ mv file1.txt are_you_learning.txt
samir@NEXUS:~/linux_lab/lab1$ ls
are_you_learning.txt doc1.txt doc2.txt doc3.txt file2.txt
samir@NEXUS:~/linux_lab/lab1$ mv are_you_learning.txt /mnt/c/Users/samir/Desktop
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt file2.txt
samir@NEXUS:~/linux_lab/lab1$ |
```

E. Deletion Commands

20. Command: `rm file2.txt`

Interpretation: Remove (delete) the file named 'file2.txt'.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt file2.txt
samir@NEXUS:~/linux_lab/lab1$ rm file2.txt
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt
samir@NEXUS:~/linux_lab/lab1$ |
```

21. Command: `rm *.txt`

Interpretation: Remove all files with .txt extension in the current directory.

Output:

```
samir@NEXUS: ~/linux_lab × + ▾
samir@NEXUS:~/linux_lab/lab1$ ls
doc1.txt doc2.txt doc3.txt
samir@NEXUS:~/linux_lab/lab1$ rm *.txt
samir@NEXUS:~/linux_lab/lab1$ ls
samir@NEXUS:~/linux_lab/lab1$ |
```

22. Command: `rmdir lab1`

Interpretation: Remove directory named 'lab1'.

Output:

```
samir@NEXUS: ~/linux_lab x + v
samir@NEXUS:~/linux_lab/lab1$ cd ..
samir@NEXUS:~/linux_lab$ ls
lab1
samir@NEXUS:~/linux_lab$ rmdir lab1
samir@NEXUS:~/linux_lab$ ls
samir@NEXUS:~/linux_lab$ |
```

23. Command: `rm -rf linux_lab`

Interpretation: Forcefully remove `linux_lab` and all its contents recursively.

Output:

```
samir@NEXUS: ~ x + v
samir@NEXUS:~/linux_lab$ cd
samir@NEXUS:~$ ls
demo demo.txt linux_lab
samir@NEXUS:~$ rm -rf demo
samir@NEXUS:~$ ls
demo.txt linux_lab
samir@NEXUS:~$ rm -rf linux_lab
samir@NEXUS:~$ ls
demo.txt
samir@NEXUS:~$ |
```

F. Additional Command Options

24. Command: `ls --version`

Interpretation: Display the version information of the `ls` command.

Output:

```
samir@NEXUS: ~ x + v - □ x
samir@NEXUS:~$ ls --version
ls (GNU coreutils) 8.32
Copyright (C) 2020 Free Software Foundation, Inc.
License GPLv3+: GNU GPL version 3 or later <https://gnu.org/licenses/gpl.html>.
This is free software: you are free to change and redistribute it.
There is NO WARRANTY, to the extent permitted by law.

Written by Richard M. Stallman and David MacKenzie.
samir@NEXUS:~$ |
```


25. Command: ls --help

Interpretation: Display help information showing all available options for the ls command.

Output:

```
samir@NEXUS: ~$ ls --help
Usage: ls [OPTION]... [FILE]...
List information about the FILEs (the current directory by default).
Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

Mandatory arguments to long options are mandatory for short options too.
  -a, --all                        do not ignore entries starting with .
  -A, --almost-all               do not list implied . and ..
  --author                       with -l, print the author of each file
  -b, --escape                   print C-style escapes for nongraphic characters
  --block-size=SIZE              with -l, scale sizes by SIZE when printing them;
                                e.g., '--block-size=M'; see SIZE format below
  -B, --ignore-backups           do not list implied entries ending with ~
  -c                             with -lt: sort by, and show, ctime (time of last
                                modification of file status information);
                                with -l: show ctime and sort by name;
                                otherwise: sort by ctime, newest first
  -C                             list entries by columns
  --color[=WHEN]                colorize the output; WHEN can be 'always' (default
                                if omitted), 'auto', or 'never'; more info below
  -d, --directory               list directories themselves, not their contents
  . . . . .
```